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Project Title:	Sensorimotor Assessment and Rehabilitation Apparatus: Procedures and Equipment		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Space Biology Special Category:	None		
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Zip Code:	21205-1832	**Congressional District:**	7
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Contact Monitor:	Loerch, Linda	**Contact Phone:**	
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Flight Assignment:	NOTE: End date changed to 7/31/2013 to reflect when new PI took over the project and for clarity (Ed., 6/13/18) NOTE: Extended to 7/31/2014 per NSSC info (Ed., 4/30/13)		
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Task Description:	Long-duration flight leads to sensorimotor problems which can be critical during landing, rendezvous, and operations on other planetary surfaces. While specific sensorimotor effects have been identified, it is not known which ones have the most adverse impact, or how best to assess them and apply appropriate rehabilitation procedures. NASA's current goal in addressing this situation is to develop a means to assess sensorimotor function rapidly with a portable device, so that an astronaut can make a determination as to whether or not he or she is impaired enough to affect mission safety or success before undertaking a demanding task (piloting, landing, reentry, egress, EVA, tele-operation, etc.). Accordingly, the goal of this project is to develop a portable hand-held device that will allow a single crewmember to assess his/her sensorimotor function in no more than 20 minutes. This is to be accomplished with a judicious choice of which sensorimotor functions to assess, and careful design to obtain the maximum data in the minimum time. The device and procedures being developed to meet these requirements are based on a notebook computer, attached to a joystick and body-mounted motion-sensor units. Through a set of simple software routines, a rapid assessment of sensorimotor capabilities will be made. The final device will be small, require little power and space, and provide what is essentially a self-contained sensorimotor lab/clinic. We term this device and its embedded software the Sensorimotor Assessment and Rehabilitation Apparatus (SARA). SARA will be designed to measure, initially, the following functional behaviors that are relevant to safe and effective functioning and are also affected by long-duration space flight: 1. Vestibulo-ocular function during pitching head movements (eye movements in response to head movements, a direct measure of vestibular function and canal-otolith integration, and highly relevant for gaze stabilization). We continue to refine our method of subjective nulling of apparent visual motion during head movement (vestibulo-ocular nulling: VON) to perform this assessment. 2. Vestibulo-ocular function as mediated by the otolith organs. We now use a binocular display with lines set by the subject to appear overlaid or parallel, to measure vertical skew and disconjugate torsion, which are measures of otolith asymmetry. 3. Postural control, by measuring body sway with eyes closed. This test is enhanced by having the subject make pitching head movements. 4. Locomotion, by measuring alterations in the coupling between pitching head movements and vertical body motions during walking.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our sensorimotor assessment apparatus has recently been implemented on a tablet computer and motion sensors with Bluetooth wireless link. This provides real-time data acquisition and processing of three linear accelerometers and three rate sensors from each of three sensors. This meets the need for rapid non-invasive screening of vestibular and sensorimotor function in settings where special expertise and equipment is not available. Applications include assessment of passengers before and after parabolic and sub-orbital flight, and assessment/rehabilitation of vestibular patients away from specialized centers. We have also refined our thinking on the total range of possible uses for SARA: 1) Go / No-Go decision before demanding task -- a) After landing (moon, Mars, NEO), b) Before EVA (deep space). 2) Data on acute deficits upon return from long-duration stay on ISS -- a) Inform countermeasure priorities. 3) On board ISS -- a) Before EVA, b) Correlate assessment with performance. 4) Field use: rapid evaluation with minimal resources -- a) Centrifuge studies, b) Before/after suborbital flight. 5) Lab adaptation studies -- a) Multi-system approach, b) Determination of individual “sensorimotor signature,” c) How do systems interact during perturbations and adaptation.
	In year 1 we had made progress in implementing assessment tests on a set of hardware for this project (HP Slate 500 tablet PC computer, Opal sensors from APDM Inc.). Based on experience with this hardware, and the fact that the Slate is no longer supported and Android-based tablets are becoming more common, we made a change to our hardware platform. We have implemented the initial set of assessment tests on a Lenovo ThinkPad tablet in Java, and assembled the necessary software components for further programming in a dedicated integrated development environment (IDE). Body motion is now transduced with a set of motion sensors from Shimmer Research (self-contained linear and angular motion sensors with Bluetooth link). This integrated set of hardware and software establishes a firm foundation for further development – in fact the current implementation is almost a complete “lab in a box,” with the capability of real-time display and acquisition of data from three 3-axis accelerometers and three 3-axis rate sensors, along with the associated processing. This SARA configuration has been validated in parabolic flight (May 2012). The assessments implemented to date include: 1. Vestibulo-ocular nulling (VON). A head-mounted motion sensor controls, through a variable gain set by the subject, the motion of a dot on the display. The subject sets the gain so that, during pitching head movements, the dot appears stationary in space. This means that it is stationary on the retina, and that dot motion matches eye motion in space. In this way, any deficiency in the VOR can be determined (i.e., how much the eyes move when the head moves). 2. Vertical and torsional skew. Red and blue lines are presented on the tablet display, and the subject wears red/blue spectacles, so that each eye sees one line. The subject is instructed to set the lines so that they appear either overlaid (in the case of vertical offsets) or parallel (for torsional offsets). This provides a subjective measure of vertical and torsional skew. 3. Posture. The subject performs the VON test while standing. This provides an additional test of vestibulo-ocular function in a natural condition, while the head movements provide a perturbation to the postural system. Head and trunk sway are measured with motion sensors. 4. Locomotion. The subject walks a predetermined path, with and without head movements (yaw or pitch), while motion sensors detect head and trunk motion. 5. Subjective vertical. The subject holds one of the motion sensors and points it toward the direction of perceived “up,” with eyes closed. Mathematical integration of the motion of the hand from a known starting orientation yields the pointing direction.

Task Progress:

Future tests to be considered for implementation include:

1. Manual control. Eye-hand tracking, following a moving point on the display with a joystick controller.

2. Vertical writing, an otolith assessment based on latent righting reflex.

3. Saccade targeting, or alternative assessment of spatial reference frame.

Lab tests – multi-system adaptation

In tests performed with the previous hardware/software version (laptop computer, APDM Opal sensors), we tested the ability of the apparatus and procedures to track sensorimotor changes during adaptation to a vestibulo-ocular perturbation. Subjects wore magnifying lenses for three hours, during which VOR, posture, and locomotion were assessed every 30 minutes. Postural sway was evaluated with 30 seconds of standing on foam, and locomotion with tandem walking. Both were performed with the head still and with sinusoidal pitching (~2 Hz). Between tests, subjects engaged in normal activities.

On donning the lenses, VOR gain was miscalibrated by more than 15%, and body sway was increased in both the posture and locomotion tests ($p<0.05$). As adaptation progressed, VOR gain adjusted to that required by the lenses ($p<0.01$), although complete adaptation did not occur over the course of the experimental trials. Body sway during standing and walking returned to baseline within the three-hour experiment.

On removal of the lenses, negative after-effects were seen: VOR gain was elevated and postural sway increased. Throughout adaptation, VOR gain was correlated with pitch and roll postural sway ($p<0.05$), although recovery was faster for posture and locomotion. The VOR perturbation invoked adaptive alteration of multiple sensorimotor systems, which are coupled but have distinct time courses, and are different between subjects.

This set of trials overall goal is to determine not only the range of adaptive changes but also couplings and interactions among the sensorimotor systems during adaptation (a “sensorimotor signature”). To this end, our initial analyses show the time course of adaptive adjustment of the VOR, posture, and locomotion over the course of the experiment. The time course of adaptation is different in different subjects. In two sample cases, VOR gain monotonically improves toward a compensatory level for each person. However, one subject shows a steady increase in body sway while walking, while the other shows an initial increase followed by a rapid decline to near-baseline levels. Postural sway likewise exhibits different time courses for each subject. Better understanding of the coupling of these different adjustments in a given person, and why different people choose different adaptive “programs,” is one of the ultimate scientific goals of this project.

More detailed examination can be seen in scatterplots of body sway while standing during pitching of the head, one axis against another. This shows coupling of motion in the two directions, the range of motion, and outliers which would indicate possible loss of stability. As with the time-course data, two sample subjects show different patterns of adaptive adjustment. One exhibits an initial large increase in fore-aft sway, which steadily decreases, transiently increases, and decreases again. This might indicate a steady improvement in postural control until the point is reached when the subject feels sufficiently adapted in the other systems to relax control (yielding an increase in sway) in order to make more natural movements that explore the limits of normal sway. Another subject has an initial small increase in sway which quickly reverts back to near-baseline levels and stays there. While it is clear that different subject exhibit different strategies, crucial insights will be gained when we determine better ways to explore the interaction between posture, locomotion, and vestibulo-ocular control, along different axes and over different time courses.

Although useful, scatterplots have a shortcoming in that they do not directly represent temporal flow. Another method of examining these data, which corrects this deficit, is through recurrence plots. Envision a state space (in the sense of state variables), in which fore-aft sway is plotted along one axis and lateral sway along the other axis (these two sways are considered as state variables). The resulting trajectory will meander in the state space, sometimes returning to places where it has been before (points of stability where sways in both directions are coupled at certain values) and sometimes wandering far afield. A recurrence plot is formed by creating time axes on the abscissa and ordinate. Each time the trajectory revisits itself (i.e., each time it returns at time t2 to a point where it had previously been at t1, within a certain tolerance), a point is plotted on the recurrence plot at (t1,t2). This captures the temporal structure of the sway.

Scatterplots for a sample subject do not show much change over time with adaptation, while the recurrence plots tell a different story. The timing of outliers is readily apparent as the presence of an empty gap with little recurrence. Conversely, restricted motion as with a stable point or high stiffness shows as a dense region. This is one example of the analytical methods that we are pursuing in attempting to understand the temporal dynamics of adaptation. It can naturally be extended to multiple parameters, allowing us to examine interactions between systems over the course of adaptation.

Parabolic flight testing I

During flights in late 2011, we demonstrated the ability of the apparatus to make measurements rapidly in a demanding environment. These tests made use of a previous version of the hardware (laptop computer, head-mounted display, Opal sensors from APDM). Measurements included pitch VOR, skew and torsion, SVV, posture, and locomotion. Changes in pitch VOR with g level were detected as expected: an increase in gain initially in high g, and a decrease in low g. Difficulties in postural stability were also measured and quantified after flight.

Parabolic flight testing II

During flights in May 2012, we ran the first dedicated trials with the new SARA hardware and software as described above. Data analysis is currently ongoing. However, we note that this platform provided stable data acquisition, with an intuitive subject interface for stimulus control and reporting. Several shortcomings were noted which will be corrected before the next set of flights (later in 2012 or 2013); these include screen glare, the ability to skip trials, and better control of viewing distance for VON testing. Future flights will determine if the measurements made with this apparatus are sensitive enough to detect changes over the course of adaptation to parabolic flight on consecutive days.

Bibliography Type: Description: (Last Updated: 01/12/2026)

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